

Here's a detailed overview of **Session 8: Virtualization & Clustering**, followed by **40 MCQs** (10 Easy, 20 Medium, 10 Hard), each with 4 options, correct answer, and explanation supported by citations.

☒ Theory Summary: Virtualization & Clustering

1. Introduction to Virtualization

Virtualization creates virtual machines (VMs) that emulate hardware, allowing multiple OS instances to run concurrently on physical hosts, dramatically improving resource utilization ([reddit.com](https://www.reddit.com/r/virtualization/)).

2. Hypervisor Types

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Type ☒ 1 (Bare-metal): Installs directly on host hardware; offers high performance and security. Examples: VMware ESXi, KVM, Hyper-V ([geeksforgeeks.org](https://www.geeksforgeeks.org/hypervisor-types/)).

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Type ☒ 2 (Hosted): Runs atop a host OS like Windows or macOS; easier setup, but incurs performance overhead. Examples: VirtualBox, VMware Workstation ([tecadmin.net](https://www.tecadmin.net/hypervisor-types/)).

3. Virtualization Modes

- **Hardware (Full) Virtualization:** Guest OS runs unmodified thanks to hypervisor and CPU support (e.g. Intel VT-x) (en.wikipedia.org).
- **Paravirtualization:** Requires OS modification for hypercalls; improves performance (reddit.com).
- **Hardware-Assisted Virtualization:** Leverages CPU-level features for better isolation and efficiency .
- **OS-Level Virtualization:** Containers share the host kernel (e.g. Docker, LXC) (reddit.com).
- **Cloning, Snapshot, Templates:**
 - **Snapshots** capture VM state at a point in time (coreskills.interviewgemini.com).
 - **Cloning** duplicates full or linked VMs.
 - **Templates** are golden images for rapid deployment.

4. Cluster Architecture & Requirements

A **cluster** links multiple nodes (physical or virtual) to provide load balancing, high availability (HA), failover, and scalability. Key requirements: shared storage, heartbeat monitoring, low-latency networking, and consistent configurations.

☒ Easy MCQs (10)

1.

****Type☒ 1hypervisors run on:****

- A) Host OS
- B) Physical hardware
- C) Containers
- D) Network switch

Answer: B

Explanation: Bare-metal architecture, no host OS (stormagic.com, geeksforgeeks.org).

2.

****Type☒ 2hypervisors run as:****

- A) Host OS apps
- B) Firmware
- C) Kernel
- D) BIOS

Answer: A

Explanation: Hosted on a guest OS (servercheap.com, tecadmin.net, techtarget.com).

3.

****Snapshot is used to:****

- A) Resize VM
- B) Capture VM state
- C) Backup BIOS
- D) Mount ISO

Answer: B

Explanation: Saves state to restore later (en.wikipedia.org, en.wikipedia.org).

4.

Paravirtualization requires:

- A) Unmodified guest OS
- B) OS modification
- C) Firmware flash
- D) No drivers

Answer: B

Explanation: Requires hypercall-aware OS (en.wikipedia.org).

5.

****Hardware-assisted virtualization uses:****

- A) GPU
- B) CPU features
- C) SSD
- D) BIOS chips

Answer: B

Explanation: Intel VT-x / AMD-V support (reddit.com, en.wikipedia.org, medium.com).

6.

OS-level virtualization uses:

- A) Hypervisor
- B) Shared kernel
- C) Dedicated hardware
- D) BIOS emulation

Answer: B

Explanation: Containers share the host kernel ([reddit.com](https://www.reddit.com), [reddit.com](https://www.reddit.com), [techtarget.com](https://www.techtarget.com), en.wikipedia.org).

7.

****Cloning duplicates a VM:****

- A) Single file
- B) Entire VM
- C) CPU only
- D) BIOS

Answer: B

8.

Template is a VM used for:

- A) Regular compute
- B) Rapid deployment
- C) Network tests
- D) BIOS update

Answer: B

9.

Cluster provides:

- A) Single point failure
- B) Scalability & failover
- C) BIOS control
- D) No networking

Answer: B

10.

Snapshots are not a replacement for:

- A) Full backups
- B) VM clone
- C) Templates
- D) Containers

Answer: A

☒ Medium (20)

11.

Type ☒ 1 offers:

- A) Low security
- B) High performance
- C) No hypervisor
- D) Shared kernel

Answer: B (reddit.com, tecadmin.net, reddit.com, techtarget.com, stormagic.com, blog.purestorage.com, coreskills.interviewgemin.com, phoenixnap.com).

12.

Type ☒ 2 drawbacks include:

- A) Direct hardware access
- B) Host OS overhead
- C) High security
- D) Bare-metal installs

Answer: B (tecadmin.net, servercheap.com).

13.

Full virtualization means:

- A) Unmodified guest OS
- B) Paravirtualized OS
- C) Host OS present
- D) BIOS sharing

Answer: A (en.wikipedia.org, stormagic.com).

14.

Paravirtualization uses:

- A) Emulation drivers
- B) Hypercalls
- C) BIOS image
- D) Containers

Answer: B .

15.

Hardware-assisted adds:

- A) No acceleration
- B) CPU support
- C) BIOS firmware
- D) Hypercall only

Answer: B (en.wikipedia.org).

16.

Snapshot overhead:

- A) None

- B) Storage usage
- C) No performance change
- D) Kernel patch needed

Answer: B .

17.

Template vs snapshot:

- A) Template is golden image
- B) Snapshot is golden image
- C) Template only for containers
- D) No difference

Answer: A

18.

Cloning type: linked clone:

- A) Independent copy
- B) Dependent on parent
- C) Not used
- D) BIOS only

Answer: B

19.

Cluster needs:

- A) Shared storage
- B) Single node
- C) Offline access
- D) No replication

Answer: A

20.

Failover sets:

- A) Single VM
- B) Mirror operation
- C) Only snapshot
- D) OS-level only

Answer: B

21.

Load balancing clusters use:

- A) Manual switch
- B) Distribute requests
- C) Only one node
- D) Backup only

Answer: B

22.

Heartbeat monitoring:

- A) Tracks node health
- B) Storage feature
- C) BIOS reset
- D) Snapshot timing

Answer: A

23.

Nested virtualization:

- A) VM inside VM
- B) BIOS update
- C) Cloning method
- D) Unused

Answer: A

24.

Paravirtualization is faster than full because:

- A) No hypercalls
- B) Reduced emulation
- C) Host OS removal
- D) Kernel bypass

Answer: B (en.wikipedia.org, dev.to, en.wikipedia.org, reddit.com, en.wikipedia.org, blog.purestorage.com).

25.

Type 1 security:

- A) Large attack surface
- B) Minimal OS
- C) Host OS dependent
- D) Not suitable for enterprise

Answer: B (phoenixnap.com, stormagic.com, dev.to).

26.

Type 2 best for:

- A) Enterprise server
- B) Dev/test
- C) Cloud provider
- D) Bare-metal only

Answer: B (blog.purestorage.com, aws.amazon.com).

27.

Container vs VM:

- A) Share kernel

B) Full emulation
C) Requires snap
D) Hypervisor needed
Answer: A ([reddit.com](#)).

28.

Snapshot RAM state:

- A) Included
- B) CPU only
- C) Kernel only
- D) Not possible

Answer: A ([reddit.com](#), [en.wikipedia.org](#)).

29.

****Full virtualization supported by hardware** like:**

- A) BIOS upgrade
- B) Intel VT-x
- C) Disk partition
- D) Template only

Answer: B ([en.wikipedia.org](#), [en.wikipedia.org](#), [techtarget.com](#)).

30.

Paravirtualization requires:

- A) No OS changes
- B) Guest OS support
- C) Snapshot drivers
- D) BIOS layer

Answer: B ([phoenixnap.com](#), [en.wikipedia.org](#), [geeksforgeeks.org](#), [stormagic.com](#)).

☒ Hard (10)

31.

Type☒ 1hypervisors avoid:

- A) Host OS overhead
- B) Hardware access
- C) Bare-metal
- D) Templates

Answer: A (stormagic.com).

32.

****Type☒ 2hypervisors performance** degraded due to:**

- A) Direct hardware access
- B) Host OS layer
- C) Kernel elimination
- D) BIOS swap

Answer: B (tecadmin.net, reddit.com).

33.

****Paravirtualization historically faster** than full but now:**

- A) Always slower
- B) Full + PV drivers faster
- C) No longer used
- D) Only on ARM

Answer: B (reddit.com).

34.

****Cluster shared storage must be**:**

- A) NAS or SAN
- B) Local disk only
- C) USB only
- D) Not needed

Answer: A

35.

Cluster networking needs <1ms latency for:**

- A) Desktop
- B) HA sync
- C) Template use
- D) Snapshot failover

Answer: B

36.

Hypercall example in Xen:

- A) PV
- B) Hyperthreading
- C) VirtualBox
- D) BIOS

Answer: A (blog.purestorage.com, en.wikipedia.org).

37.

Snapshot stack complexity:

- A) Simple
- B) Overlay layers
- C) No impact

D) Template only

Answer: B (en.wikipedia.org, reddit.com).

38.

****Hardware-assisted techniques** benefit:**

A) Mouse drivers

B) Nested virtualization

C) BIOS only

D) Clone speed

Answer: B (en.wikipedia.org, en.wikipedia.org, techtarget.com).

39.

Cluster consistency needs:

A) Manual sync

B) Configuration management

C) Snapshots only

D) Host reboot

Answer: B

40.

Container virtualization is:

A) Hardware-based

B) OS-level

C) Snapshot-only

D) Hypervisor-based

Answer: B (en.wikipedia.org, phoenixnap.com, reddit.com).

Let me know if you need answers explained or are ready for Session 9!